

BUSINESS REQUIREMENTS DOCUMENT (BRD)

AI-Powered Travel & Hospitality Ecosystem Platform

Prepared For

Prayogbharti Foundation

Version

1.0

Document Classification

Confidential

Date

June 2026

1. Executive Summary

Prayogbharti Foundation proposes to develop a comprehensive AI-driven Travel & Hospitality Ecosystem Platform designed to transform the travel industry through intelligent automation, digital hotel onboarding, inventory management, dynamic pricing, AI-powered trip planning, and personalized travel experiences.

The platform will serve as a centralized digital marketplace connecting travelers, hotels, travel agents, tour operators, corporate customers, transportation providers, and tourism stakeholders through a unified technology ecosystem.

The solution aims to compete with and surpass existing Online Travel Agencies (OTAs) such as MakeMyTrip, Goibibo, Agoda, Booking.com, Expedia, and Airbnb by leveraging Artificial Intelligence, Machine Learning, Predictive Analytics, and Multi-Agent AI Architecture.

2. Business Objectives

Primary Objectives

- Create a unified travel marketplace

- Digitize hotel onboarding and inventory management
- Enable AI-assisted travel planning
- Increase hotel occupancy rates
- Improve traveler experience through personalization
- Reduce operational costs through automation
- Generate new revenue opportunities for hospitality partners
- Provide end-to-end travel management capabilities

Strategic Objectives

- Become a leading AI-native travel platform
 - Support domestic and international tourism
 - Enable MSME hotels to digitize operations
 - Promote sustainable tourism initiatives
 - Create employment opportunities in tourism and hospitality sectors
-

3. Project Vision

To establish a next-generation intelligent travel platform that enables travelers to plan, book, manage, and experience journeys through AI-powered recommendations while enabling hospitality businesses to maximize revenue through automated inventory, pricing, and customer engagement management.

4. Scope of Work

The project includes development of:

Traveler Portal

- Destination Discovery
- Hotel Search
- Flight Search
- Package Booking
- Activity Booking
- Visa Assistance
- Travel Insurance
- AI Travel Planner
- AI Travel Assistant

Hotel Partner Portal

- Property Registration
- Inventory Management
- Booking Management

- Revenue Dashboard
- Pricing Management
- Analytics Dashboard

Corporate Travel Portal

- Employee Travel Requests
- Approval Workflow
- Travel Policy Management
- Budget Monitoring
- Expense Tracking

Travel Agent Portal

- Customer Management
- Group Bookings
- Commission Tracking
- Quotation Generation

Administration Portal

- User Management
 - Hotel Approval
 - Booking Monitoring
 - Revenue Analytics
 - Content Management
 - Fraud Detection
-

5. Stakeholders

Internal Stakeholders

- Prayogbharti Foundation
- Product Management Team
- Technology Team
- Operations Team
- Marketing Team

External Stakeholders

- Hotels
- Resorts
- Homestays
- Travel Agencies
- Airlines
- Transportation Providers

- Tour Operators
 - Corporate Customers
 - Individual Travelers
-

6. Functional Requirements

FR-001 Hotel Registration & Onboarding

Description

Hotels shall be able to register digitally through a self-service onboarding portal.

Features

- Hotel Profile Creation
- KYC Verification
- GST Verification
- PAN Validation
- Bank Account Verification
- Property Documentation Upload
- Geo-location Mapping
- Room Configuration Setup
- Amenity Selection

AI Features

- OCR Document Extraction
 - Address Validation
 - Duplicate Property Detection
 - Property Classification
 - Content Generation
-

FR-002 Inventory Management

Description

Hotel partners shall manage room inventory through centralized inventory management.

Functions

- Room Creation
- Availability Management
- Occupancy Tracking
- Maintenance Blocking

- Reservation Calendar
- Bulk Inventory Upload

AI Capabilities

- Occupancy Forecasting
 - Demand Prediction
 - Inventory Optimization
 - Cancellation Forecasting
-

FR-003 Dynamic Pricing Engine

Description

The platform shall automatically recommend room pricing.

Inputs

- Historical Booking Data
- Competitor Pricing
- Seasonal Demand
- Local Events
- Occupancy Rates

Outputs

- Recommended Price
 - Dynamic Discounts
 - Surge Pricing
 - Revenue Optimization Suggestions
-

FR-004 Booking Engine

Capabilities

- Hotel Booking
- Flight Booking
- Package Booking
- Multi-City Itinerary Booking
- Group Reservations
- Corporate Reservations

Features

- Secure Payment Processing

- Booking Confirmation
 - Cancellation Management
 - Refund Processing
-

FR-005 AI Trip Planner

Description

Travelers shall interact with an AI-powered travel assistant capable of generating personalized itineraries.

Sample User Query

"Plan a 7-day family trip to Kerala within ₹80,000."

AI Output

- Flights
 - Hotels
 - Local Transportation
 - Activities
 - Attractions
 - Estimated Budget
 - Daily Schedule
-

FR-006 AI Travel Concierge

Features

- Conversational Booking
- Travel Guidance
- Visa Information
- Emergency Support
- Weather Alerts
- Local Recommendations
- Booking Modifications

Channels

- Website
 - Mobile Application
 - WhatsApp
 - Voice Assistant
-

FR-007 Review & Reputation Management

Functions

- Review Collection
- Rating Management
- Complaint Management
- Sentiment Analysis
- Review Moderation

AI Features

- Fake Review Detection
 - Sentiment Scoring
 - Complaint Categorization
-

FR-008 OTA Channel Management

Integrations

- MakeMyTrip
- Agoda
- Booking.com
- Expedia
- Airbnb
- Goibibo

Functions

- Inventory Synchronization
 - Rate Synchronization
 - Reservation Synchronization
-

FR-009 Revenue Management System

KPIs

- ADR
- RevPAR
- Occupancy Percentage
- Revenue Forecast
- Cancellation Rate

AI Features

- Revenue Forecasting
 - Upselling Recommendations
 - Pricing Optimization
-

7. AI Architecture

AI Agents

Travel Planning Agent

Responsible for:

- Destination Research
- Itinerary Generation
- Travel Recommendations

Hotel Onboarding Agent

Responsible for:

- Document Verification
- Data Extraction
- Property Categorization

Pricing Agent

Responsible for:

- Demand Forecasting
- Dynamic Pricing

Customer Support Agent

Responsible for:

- Traveler Support
- Issue Resolution
- FAQs

Fraud Detection Agent

Responsible for:

- Fraud Monitoring

- Suspicious Transaction Detection
 - Fake Booking Detection
-

8. Non-Functional Requirements

Performance

- Response Time < 3 Seconds
- Concurrent Users: 100,000+
- Availability: 99.95%

Scalability

- Multi-Country Expansion
- Multi-Language Support
- Multi-Currency Support

Security

- ISO 27001 Compliance
- PCI DSS Compliance
- DPDP Act Compliance
- GDPR Compliance

Security Controls

- MFA
 - Role-Based Access Control
 - Data Encryption
 - Audit Logs
 - SIEM Integration
 - WAF Protection
-

9. System Integrations

Payment Gateways

- Razorpay
- PayU
- Stripe
- PayPal

Maps

- Google Maps
- Mapbox

Communication

- WhatsApp Business API
- SMS Gateway
- Email Gateway

Travel APIs

- Amadeus
 - Travelport
 - Sabre
 - Skyscanner
-

10. Technology Architecture

Frontend

- ReactJS
- NextJS
- Flutter

Backend

- NodeJS
- Java Spring Boot

Database

- PostgreSQL
- MongoDB
- Redis

Search

- Elasticsearch

Cloud

- Microsoft Azure

AI Framework

- OpenAI
- Gemini
- LangGraph
- CrewAI

DevOps

- Docker
 - Kubernetes
 - Azure DevOps
-

11. Reporting & Analytics

Executive Dashboard

- Total Bookings
- Revenue
- Active Hotels
- Customer Acquisition

Hotel Dashboard

- Occupancy Rate
- Revenue Trends
- Booking Sources

AI Insights

- Demand Forecast
 - Market Trends
 - Revenue Opportunities
-

12. Project Phases

Phase 1 – Foundation Platform

Duration: 6 Months

Deliverables:

- Hotel Onboarding

- Inventory Management
 - Booking Engine
 - Basic AI Planner
-

Phase 2 – Marketplace Expansion

Duration: 6 Months

Deliverables:

- Dynamic Pricing
 - Channel Manager
 - Corporate Travel Module
-

Phase 3 – AI Ecosystem

Duration: 12 Months

Deliverables:

- Multi-Agent AI Platform
 - Predictive Analytics
 - Revenue Optimization
 - Travel Marketplace
-

13. Expected Benefits

For Travelers

- Personalized Travel Experiences
- Faster Trip Planning
- Cost Optimization
- Enhanced Customer Support

For Hotels

- Increased Occupancy
- Automated Operations
- Revenue Optimization
- Better Market Visibility

For Prayogbharti Foundation

- Sustainable Revenue Generation
 - National Tourism Enablement
 - Employment Generation
 - Technology Leadership
-

14. Success Metrics

- 10,000+ Hotels Onboarded
 - 1 Million Registered Users
 - 500,000 Annual Bookings
 - 30% Reduction in Customer Service Costs
 - 20% Increase in Hotel Revenue through AI Recommendations
 - 99.95% Platform Availability
-

15. Conclusion

The proposed AI-Powered Travel & Hospitality Ecosystem will establish Prayogbharti Foundation as a technology-driven enabler of the tourism and hospitality sector. By combining intelligent automation, AI-driven travel planning, dynamic inventory management, and predictive analytics, the platform will create a scalable and sustainable ecosystem serving travelers, hotels, and tourism stakeholders across India and international markets.